CAPSTONE PROJECT

Autonomous Data-Analysis Agent

Ask a question about your data in plain English. Get a verified answer, the code behind it, and the rows it used.

Team: [your names] · Supervisor: [name] · October 2026



Chat-style AI can't be trusted with your data

42?

Made-up numbers

A language model can state a figure it never actually computed from the data.

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No evidence

You can't see the code or the rows behind the answer, so you can't check it.

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Privacy risk

Pasting raw customer data into an outside service exposes names and emails.

Our goal: answers that are computed, shown, checked, and private.

One question in, a verified answer out

THE USER ASKS

“How many orders came from customers in the West region?”

Uploads: orders.csv and customers.csv

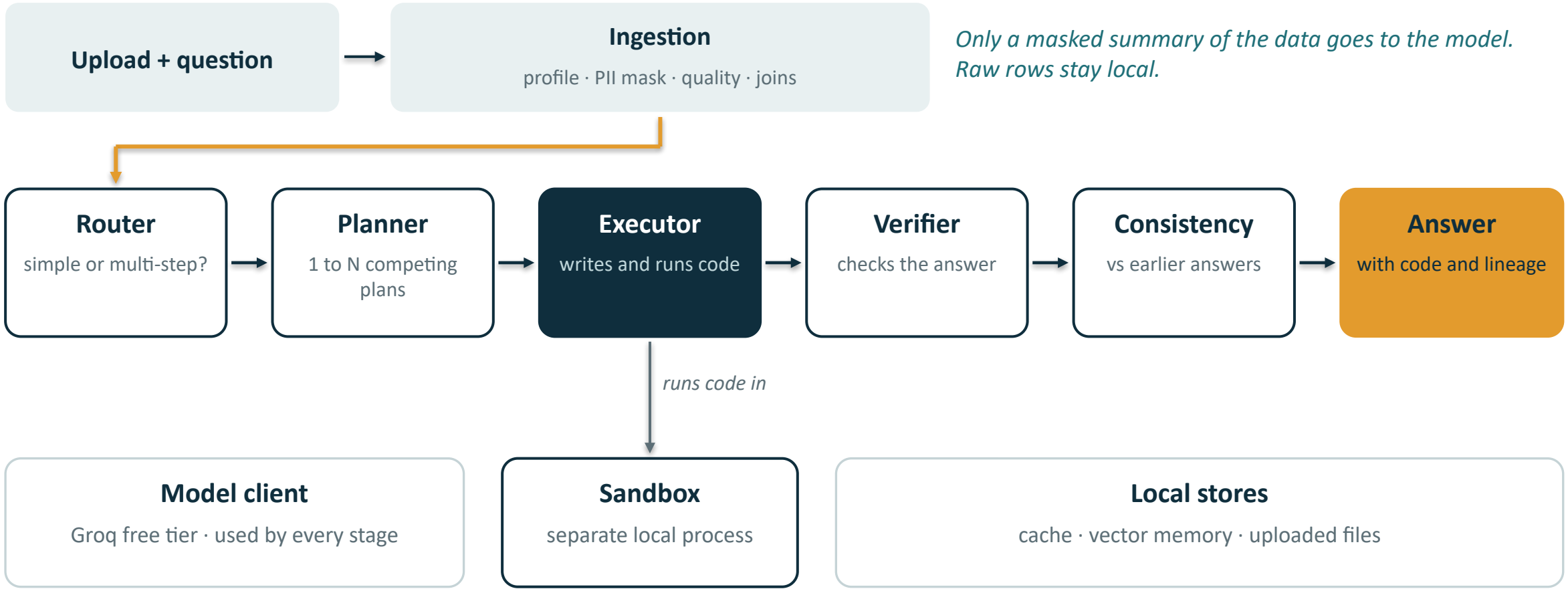
The two files are linked automatically on customer_id.

No code or SQL needed from the user.

THE AGENT RETURNS

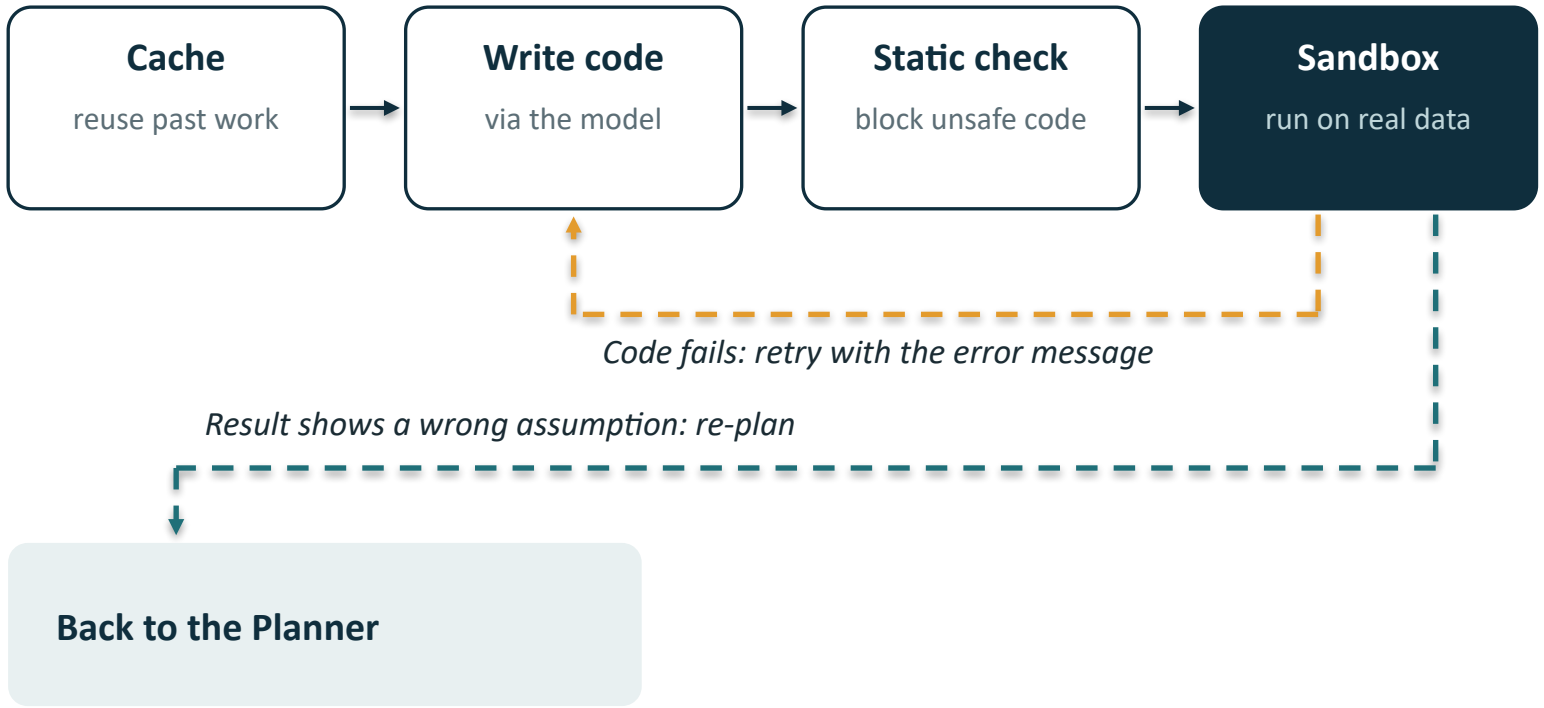
Answer	**11 orders**
Code	The exact pandas code that ran, shown and exportable as a notebook
Lineage	Which columns were used: orders.customer_id, customers.region
Confidence	A bootstrap range when the data has quality issues
Scope note	A warning when the question asks more than the data can support

A five-stage pipeline, built as a LangGraph state graph



The agent catches and fixes its own mistakes

Every step of every plan goes through this loop.



3 retries per step, each one shown the previous error

2 re-plans per question when a result breaks the plan

N competing interpretations run in parallel, each in its own sandbox

Private data is masked, and code runs sandboxed

WHAT THE MODEL SEES

Column	Real value	Sent to the model
name	`Alice Johnson`	`A**********n`
email	`alice.johnson@example.com`	`a***…***m`
region	`West`	`West`

Name and email columns are masked before any model call.

Calculations still use the real values, so answers stay exact.

Region is not personal data, so it passes through unchanged.

SANDBOX SAFEGUARDS

Separate process

Generated code never runs inside the app itself

Static check

Code is inspected before it runs, unsafe calls are refused

No network

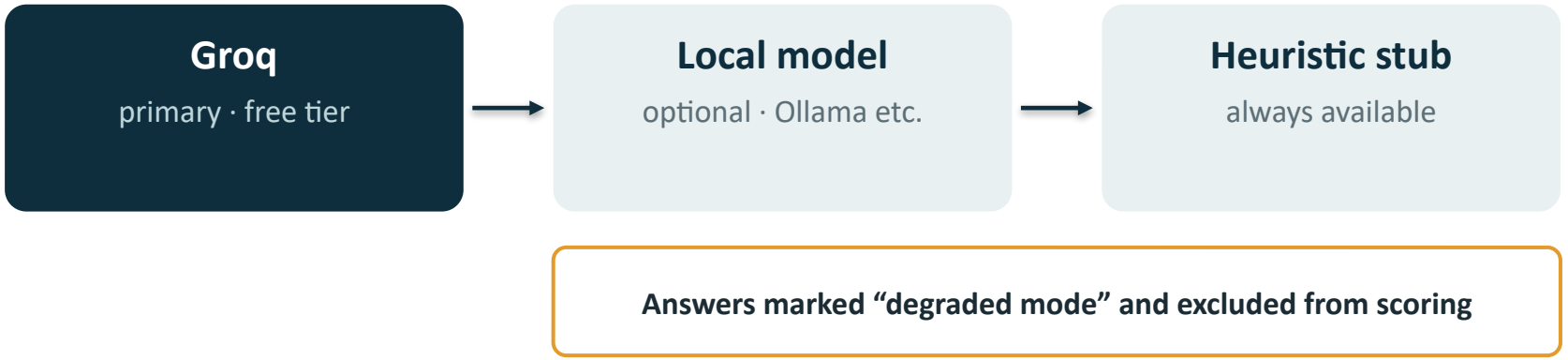
Network access is blocked inside the sandbox

Hard limits

20-second timeout and 512 MB memory cap, enforced from outside

Zero cost, with open fallbacks when a model fails

IF A MODEL TIER FAILS 3 TIMES IN 2 MINUTES, THE NEXT ONE TAKES OVER



MODELS PER STAGE

Router, code writer	`gpt-oss-20b`	fast, cheap decisions
Planner, verifier	`gpt-oss-120b`	where reasoning quality matters most

$0

model cost per question

Groq's free tier hosts open models. The project tracks tokens and latency instead of dollars.

Live demo

1

Upload both sample files

orders.csv and customers.csv. Watch the name and email columns get flagged as personal data.

2

“How many orders came from customers in the West region?”

Shows the agent linking two files on its own. Expected answer: 11.

3

“What is the best-selling product?”

Ambiguous: by units or by revenue? Watch the planner try competing interpretations.

4

Ask a follow-up question

Shows session memory and the consistency check against the earlier answer.

5

Open the Operations tab

Model status, sandbox health, review queue and token use in one view.

7 of 7 benchmark questions answered correctly

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automated tests passing, including the real sandbox process and timeout kill

100%

benchmark pass rate on 7 scored questions

BENCHMARK BY CATEGORY

aggregation	pass	Total number of orders
aggregation	pass	Total quantity sold
filter	pass	Orders for Widget A
join	pass	Orders from West-region customers
time series	pass	Orders placed in Q1 2023
ambiguous	degraded	Best-selling product
pii masked	pass	Unique customer regions
large data	pass	Average unit price

Run on Groq's free tier, October 2026. 8 seed questions with known answers. 1 question hit a model fallback during the run, so it was excluded from scoring even though its answer was correct.

Known limits and what we would do next

Area	Today	Next step
PII detection	Patterns and column-name rules	Trained named-entity model
Similarity search	Local hashing vectors, no network	Real embedding model
Isolation	Separate OS process, no container	Docker or VM sandbox
Benchmark	8 seed questions	50+ questions across all categories
Scale	One machine	Pool of sandboxes on Kubernetes

Thank you

We'd value your feedback on scope and on what to prioritise next.

Questions?

PROPOSED NEXT STEPS

Grow the benchmark

to 50+ questions so the pass rate is meaningful

Stronger PII detection

with a trained model instead of patterns

Container sandbox

for safe multi-user deployment